

KRANDELLE JOSH B. KALAW

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PROFESSIONAL SUMMARY

Results-driven Information Technology graduate (completed all academic and practicum requirements; graduating September 2026) and aspiring Software Developer with a strong foundation in Object-Oriented Programming, Data Structures, and Modern Web Technologies. Proven experience across the full software development lifecycle, having solely engineered a complex desktop hardware controller application using React 19 and Electron. Backed by enterprise IT infrastructure internship experience at Lufthansa Technik Philippines. Highly adaptable, available for immediate full-time employment, and eager to leverage advanced engineering practices including AI-assisted coding and robust prompt engineering workflows to build scalable software solutions.

TECHNICAL SKILLS

- **Core Languages:** JavaScript (ES6+), HTML5, CSS3, JSX, Python, SQL
- **Frameworks & Libraries:** React 19, Electron 39, Node.js, React Testing Library
- **Hardware & IoT Integration:** USB Serial Communication (serialport), GRBL Controllers, ESP32, Arduino, G-code Generation
- **Tools & Build Systems:** Git, npm, electron-builder, Create React App, cross-env, concurrently
- **Development Practices:** AI-Assisted Development & Prompt Engineering, Object-Oriented Programming (OOP), Data Structures & Algorithms, Secure IPC (contextBridge)
- **IT & Systems Operations:** Systems Administration, Network Fundamentals, Hardware/Software Troubleshooting, Helpdesk Ticket Resolution

TECHNICAL PROJECTS

LaserPix | *Desktop Laser Controller & Image Processing System (Capstone)* github.com/Krandelle/LaserPix

- Engineered a robust desktop application using React 19 and Electron 39, featuring a modern, responsive UI for real-time hardware control and pixel-perfect image processing.
- Developed a highly reliable USB serial communication layer using Node.js serialport, enabling auto-detection, dynamic baud rate management, and seamless G-code streaming to ESP32-based GRBL controllers.
- Designed dynamic hardware profiles to manage device-specific command delays and ensure precise timing synchronization between the software interface and the physical laser module.
- Secured the Electron application architecture using contextBridge and preload.js to safely expose native Node.js APIs to the React frontend without compromising system security boundaries.
- Automated the Windows NSIS installer packaging workflow using electron-builder, optimizing build scripts with cross-env and concurrently for rapid, unified development and deployment.

WORK EXPERIENCE

Lufthansa Technik Philippines | *IT Intern* March 2026 – June 2026

- Supported enterprise IT operations, hardware troubleshooting, and critical systems maintenance across corporate and hangar departments, ensuring optimal system operational performance.
- Assisted in complex network configurations and system administration tasks within a large-scale aviation IT infrastructure, gaining deep exposure to enterprise-level system integration.

- Resolved technical helpdesk tickets efficiently, utilizing analytical problem-solving skills to maximize system uptime and productivity for cross-functional corporate teams.
- Managed meticulous technical documentation and inventory logs for active workstations, ensuring highly accurate tracking of critical enterprise IT assets and system configurations.

Freelance Remote Contracts | *Video Editor & Multimedia Specialist*

2023 – 2025

- Managed comprehensive end-to-end digital workflows, editing, mixing, and refining high-quality video/audio content (including YouTube tutorials and podcasts) while strictly meeting aggressive project deadlines.
- Maintained organized cloud file structures, managed modern export formats, and optimized multi-threaded rendering processes to ensure high-fidelity output across diverse digital platforms.
- Communicated effectively with global clients via email and chat to gather granular technical requirements, translate creative feedback into actionable items, and ensure seamless delivery.

Remote Operations | *Virtual Assistant Intern (SHS Work Immersion)*

2021 – 2022

- Completed a highly structured corporate work immersion program, handling calendar scheduling, high-volume data entry, and core administrative workflows using Google Workspace.
- Assisted in secure document management and basic analytical reporting, cultivating strong organization, professional ethics, and crisp corporate communication skills.
- Maintained absolute confidentiality and strict data accuracy across all assigned administrative tasks while effectively supporting daily team operations.

EDUCATION

Cavite State University - Imus Campus | *Bachelor of Science in Information Technology*

Graduating Sept 2026

Status: Completed all academic coursework and enterprise practicum requirements. Awaiting formal graduation. (*Available for immediate full-time employment*)

Relevant Coursework: Object-Oriented Programming, Data Structures & Algorithms, Web Systems & Technologies, Advanced Database Management, Systems Integration & Architecture, App Development & Emerging Technologies.

St. Vincent De Ferrer College of Camarin | *Senior High School (STEM Strand)*

2020 – 2022